

Arduino-Based Gauss Meter



RONIE ADHIRAJ GHOSH

Presented here is a handheld device that can be used to measure magnetic field using a Hall-effect-based sensor. The device can be used in the physics lab for conducting various experiments on magnetism. Moreover, some upgradations can be made to develop new and advanced devices. The device shows magnetic field in CGS (centimetre-gram-second system) unit, Gauss and is updated every 250 milliseconds on the LCD screen.

Circuit and working

Fig. 1 shows the circuit diagram of Arduino-based Gauss meter. The circuit is built around Arduino Uno board (Board1), MCP3202 12-bit analogue-to-digital (A/D) converter (IC1), WSH315 Hall-effect sensor (IC2) and serial LCD (LCD1).

MCP3202 (12-bit A/D converter).

The Microchip Technology's MCP3202 is a successive approximation 12-bit A/D converter with an on-board sample-and-hold circuit. It is programmed to provide a single pseudo-differential input pair or dual single-ended inputs. Differential non-linearity (DNL) is specified at ± 1 LSB, and integral non-linearity (INL) is offered in ± 1 LSB (MCP3202-B) and ± 2 LSB (MCP3202-C) versions. Communication with the device is done using a simple serial interface compatible with SPI protocol. The device is capable of conversion rates of up to 100ksp/s at 5V and 50ksp/s at 2.7V. The MCP3202 device operates over a broad voltage range of 2.7V to 5.5V. Low-current design permits operation with typical standby and active currents of only 500nA and 375 μ A, respectively.

WSH315 (Hall-effect sensor). This ratiometric, linear Hall-effect sensor accurately tracks extremely small changes in magnetic flux density, even

those changes that are generally too small to operate Hall-effect switches. The device comes with a Hall-sensing element, linear amplifier and emitter-follower output stage integrated on a single IC to handle small analogue signals.

Arduino Uno board. Arduino is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended

for artists, designers, hobbyists and anyone else interested in creating interactive objects or environments. Arduino Uno is a board based on ATmega328 microcontroller. It consists of 14 digital input/output pins, six analogue inputs, a USB connection for programming the on-board microcontroller, power jack, an ICSP header and a reset button. It is operated with

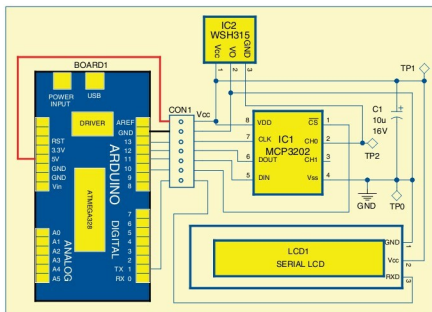


Fig. 1: Circuit diagram of Arduino-based Gauss meter

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	Output of Hall-effect sensor

PARTS LIST

Semiconductors:	
IC1	- MCP3202, 12-bit ADC
IC2	- WSH315, Hall-effect sensor
LCD1	- 16x2 serial LCD
Capacitor:	
C1	- 10 μ F, 16V electrolytic
Miscellaneous:	
BOARD1	- Arduino Uno board
CON1	- 7-pin connector

a 16MHz crystal oscillator and contains everything needed to support the microcontroller. It is very easy to use as the user simply needs to connect it to a computer with a USB cable or power it with an AC-to-DC adaptor or battery to get started. The microcontroller on the board is programmed in Arduino programming language using Arduino development environment. Pins 10 through 13 of Board1 are connected to pins 1, 5, 6 and 7 of 12-bit ADC IC1, respectively. Pin 1 of Board1 is connected to RXD pin 3 of LCD1.

The working of the project is simple. Hall-effect sensor IC2 has 3 pins: Vcc, Ground and Vout. The Vout of